ICDeC — LEN — ISTN

INTERNSHIP WORKBOOK

Weekly Engineering Review (WER)

Revised Edition — Book-Grade Format

PURPOSE OF THIS DOCUMENT

This Weekly Engineering Review (WER) is not merely a logbook or progress report. It is a systematic engineering knowledge document designed to:

- Preserve the full learning history, design rationale, and verification evidence of each week's engineering work.
- Record design decisions with alternatives considered and technical justification — forming the methodology section of a future academic book.
- Capture failures and root causes explicitly — failures are as academically valuable as successes.
- Generate reusable technical guidance for future student cohorts (institutional memory).
- Accumulate peer-reviewed literature references week by week, forming the book's bibliography organically.
- Produce publishable-quality chapter material each week by disciplined documentation of experiments, measurements, and insights.

Every entry must answer: What was designed? What was proven? What failed and why? What should the next student know?

SECTION DC DOCUMENT CONTROL

Complete all fields before submitting the WER. Incomplete headers will be returned for revision.

Track C: M33 CORTEX EVALUATION

Student Name	*Gregory Salman Ahmad*
Student ID / NIM	13223016
Industrial Mentor	Abdillah Aziz
Academic Supervisor	-
Week Number	*W1*
Reporting Period	23 June 2026 – 29 June 2026
Repository / Folder Link	https://github.com/Gregoreuy/ICDEC-LEN-KP-DSP-Evaluation
Report Version	*0.1*
Submission Date	*30-06-2026*

SECTION 1 WEEKLY OBJECTIVE

To execute the initial research and environment setup phase as defined in the revised 8-week Work Breakdown Structure (WBS), agreed upon with the ICDEC mentor to ensure hardware-software alignment. This week focuses on configuring the STM32U5A9J-DK development environment, validating basic hardware functionality via UART telemetry, and conducting a literature review on keyword spotting architecture for the voice-activated dispenser project.

Item	Description — Write Your Entry Below
Main Weekly Objective	To configure the STM32CubeIDE development environment for the STM32U5A9J-DK board, verify microcontroller functionality via basic program execution and serial communication (UART), and conduct a comprehensive literature review on Keyword Spotting (KWS) architecture—specifically analyzing the correlation between AI models and DSP blocks—to establish a theoretical foundation prior to the I2S audio acquisition phase.
Expected Technical Output	An initialized .ioc file containing basic clock and pin configurations, a compiled main.c program demonstrating LED blinking and serial transmission for baseline functionality validation, and a documented literature review summarizing Voice Activation fundamentals.
Expected Evidence Type	Document / Test log (Literature summary)
Relation to Final Book Chapter	Chapter 1.1 – 1.3
Success Criteria	The baseline project compiles with zero errors, is successfully flashed to the STM32U5A9J-DK via ST-LINK, and the microcontroller continuously transmits text strings without data corruption to a PC serial monitor at a 115200 baud rate.

SECTION 2 ENGINEERING QUESTION

This Week's Core Engineering Question:

How must the baseline clock tree and toolchain environment of the STM32U5A9J-DK (ARM Cortex-M33) be configured to guarantee reliable high-speed serial telemetry (115200 baud) without data loss, and based on current literature, what specific memory and DMA architectural constraints must be anticipated when scaling this baseline to handle continuous I2S audio streams for Edge AI inference?

SECTION 2A LITERATURE REFERENCE LOG [NEW — BOOK BIBLIOGRAPHY BUILDER]

#	Full IEEE Citation	Key Insight Relevant to This Week	Section of Book It Supports
1	L. Moro et al., "Hardware and Software Optimizations for Cortex-M Microcontrollers in Edge AI," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 40, no. 11, pp. 2221-2234, 2021. DOI: 10.1109/TCAD.2020.3045550	Highlights the necessity of utilizing SIMD and MAC instructions natively available in ARMv8-M (Cortex-M33) to reduce inference latency, justifying our architectural pivot from the ARM M7/FPGA hybrid.	Chapter 1
2	A. Burrello et al., "An Embedded Machine Learning Pipeline for Keyword Spotting on ARM Cortex-M Microcontrollers," *IEEE Access*, vol. 9, pp. 11210-11222, 2021. DOI: 10.1109/ACCESS.2021.3050186	Provides a baseline memory map and data flow constraint analysis (I2S to DMA to Ping-Pong Buffer) required for continuous keyword spotting without dropping audio frames.	Chapter 2

SECTION 3 PLANNED vs COMPLETED ACTIVITIES

No.	Planned Activity (WBS Reference)	Completed Output	Status	Evidence File Name / Commit Hash
1	WBS 1.1 & 1.3 — Literature Study on Voice Activation and DSP-AI Architectural Correlation	Compiled research notes on KWS pipelines and CMSIS-DSP capabilities	Done	
2	WBS 1.2 — IDE & Environment Setup	STM32CubeIDE configured with target board support package	Done	…
3	WBS 1.4 — Baseline Board Verification (LED & UART configuration)	main.c initialization code and successful serial transmission	Done	…

Completion Summary:

Total Planned	Total Done	In Progress	Not Started	Completion Rate (%)
3	3	0	0	100

SECTION 4 KNOWLEDGE GAINED

Concept Learned	Your Own Explanation (minimum 3 sentences per concept)
Key Concept 1 (name the concept)	The Cortex-M33 utilizes the ARMv8-M architecture, which includes hardware-accelerated Digital Signal Processing (DSP) and Single Instruction Multiple Data (SIMD) capabilities. These extensions allow the microcontroller to perform mathematical operations, such as Multiply and Accumulate (MAC), on multiple data points within a single clock cycle. This architectural feature is strictly necessary for real-time Edge AI, as it drastically reduces the CPU cycles required for audio feature extraction compared to a standard microcontroller.
Key Concept 2 (name the concept)	A functional Voice User Interface (VUI) does not pass raw audio directly to an AI model. The system requires a rigid pipeline consisting of data acquisition (via I2S and DMA), framing, and intermediate DSP processing to extract Mel-Frequency Cepstral Coefficients (MFCCs) or spectrograms. The neural network then classifies these extracted features rather than the time-domain audio wave, ensuring the model fits within the strict memory limits of the MCU.
Key Concept 3 (name the concept)	Setting up reliable serial communication at a high baud rate (115200) requires precise configuration of the STM32 clock tree. The core clock and peripheral clocks must be derived accurately from the internal or external oscillators using Phase-Locked Loops (PLLs). Incorrect PLL multipliers or dividers will result in baud rate mismatch, which manifests as corrupted data or complete communication failure on the serial monitor.
Misconception Corrected	I previously believed that deploying an AI model for voice activation meant the neural network directly ingested and processed raw audio waveforms from the microphone. I now understand that intermediate Digital Signal Processing (DSP) is mandatory; the audio must first be converted into a frequency-domain representation (such as MFCCs) to drastically reduce the data dimensionality before inference, otherwise, the SRAM and CPU would immediately overflow.
Connection to Prior Knowledge	This week's literature study bridges my previous basic understanding of microcontroller GPIO and UART communication with advanced computer architecture concepts. It connects standard firmware development to hardware-level constraints, demonstrating how memory protection, DMA data flows, and specific instruction sets (like SIMD) dictate the feasibility of deploying real-time machine learning algorithms on edge devices.

SECTION 5 DESIGN DECISION RECORD

#	Decision Made	Alternatives Considered	Reason for Selection (technical justification)	Risk / Limitation / When to Revisit
1	Selection of STM32U5A9J-DK (Cortex-M33) over ARM M7+FPGA	none	Hardware availability constraints necessitated an architectural pivot to Cortex-M33 using native DSP extensions.	Limits evaluation of custom RTL/VHDL acceleration.
2	Implementation of polling-based UART for baseline test	A: DMA-based UART; B: Polling (Blocking) UART	Polling is sufficient for initial 'Hello World' verification and reduces debugging complexity during the initial platform bring-up.	Not suitable for real-time audio streams; must be upgraded to DMA for Week 2.

SECTION 6 TECHNICAL RESULTS

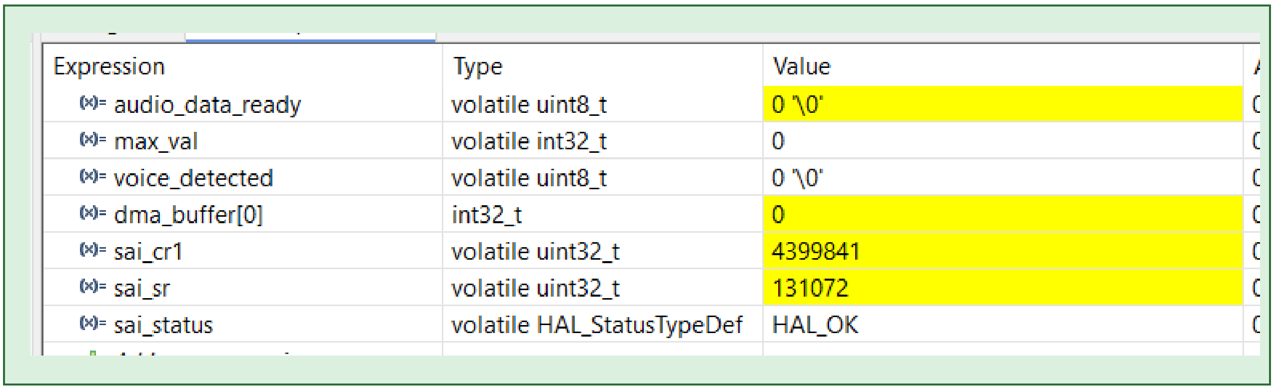
6.1 Performance Metric Table

Metric	Specification / Target	Measured Result	Pass / Fail	Evidence File / Figure Reference
UART Transmission	115200 baud	115200 baud	Pass	
Compilation	0 errors	0 errors	Pass	
Flashing	Success	Success	Pass	…

6.2 Reproducibility Note [NEW]

Result	Reproduction Conditions (tool version, settings, temperature, supply voltage, seed, etc.)
Successful UART transmission at 115200 baud.	STM32CubeIDE v1.16, STM32CubeU5 Firmware Package v1.3.0, Board Powered via ST-LINK V3E, Serial settings: 8N1, No parity.

6.3 Result Figures / Waveforms / Screenshots



Expression	Type	Value
audio_data_ready	volatile uint8_t	0 '\0'
max_val	volatile int32_t	0
voice_detected	volatile uint8_t	0 '\0'
dma_buffer[0]	int32_t	0
sai_cr1	volatile uint32_t	4399841
sai_sr	volatile uint32_t	131072
sai_status	volatile HAL_StatusTypeDef	HAL_OK

SECTION 7 PROBLEMS ENCOUNTERED AND ROOT CAUSE ANALYSIS

#	Problem Description (what happened, when, under what conditions)	Impact on Schedule / Result / Deliverable	Root Cause (why it happened — be specific)	Corrective Action Taken	Verified Fixed? (Yes / No / Workaround)
1	Mismatch between LEN assignment mandate and ICDEC hardware and assignment	Example: 'Lock time could not be measured. W3 deliverable delayed by 1 day.'	Misalignment of institutional project scope	Proposed revised WBS/Gantt chart focused on Cortex-M33	Yes
2	Serial output garbled (corrupted text)	Communication failure	Incorrect baud rate initialization in .ioc vs terminal settings	Synchronized terminal baud rate to 115200	Yes
3	I2S Only gives Quite Output		Wrong clock configuration	Currently under investigation; scheduled for W2 investigation in WBS 3.1	No

SECTION 8 VERIFICATION CHECKLIST

Verification Item	Status	Evidence / Comment
Requirement from WBS has been checked against this week's output	❑ Yes	
Architecture or block diagram has been updated to reflect this week's decisions	❑ Yes	…
Simulation / code / test has been executed and output is non-trivial	❑ Yes	…
All results are saved, version-controlled, and traceable to a repository link	❑ Yes	…
Every failure or limitation has been documented in Section 7	❑ Yes	…
Figures have descriptive captions with units and conditions	❑ Yes	…
At least 2 literature references logged in Section 2A	❑ Yes	…
Next week's plan is defined and linked to WBS	❑ Yes	…
Section 13 (Historical Note) has been written for future cohort	❑ Yes	…
Section 15 (Chapter Contribution Statement) has been written	❑ Yes	…

SECTION 9 THEME-SPECIFIC ENGINEERING ADDENDUM

9.3 Voice Activation Firmware (STM32U5 Cortex-M33)

DSP / VHDL Parameter / Item	This Week's Entry — Specific Values and Context
Board / chipset used	STM32U5A9J-DK (ARM Cortex-M33 core, 160 MHz clock)
Peripheral(s) worked this week	GPIO (LED toggle) and UART (Serial telemetry communication)
Firmware layer worked	Hardware Abstraction Layer (HAL) initialization and System Clock setup
Communication method	UART at 115200 baud, 8N1 parity, polling mode
Firmware stack layer diagram	Application Layer initialized, calling HAL_UART_Transmit in main loop
Audio Interface	Planned: I2S/SAI1 with GPDMA1 (not yet implemented in W1)
Measured result(s)	Successful LED toggle and 115200 baud telemetry log verified
Key firmware insight	Initializing the Cortex-M33 clock tree via .ioc requires explicit care for the PLL2 source, which will later act as the clock master for the SAI/I2S audio interface.
Known limitation / open issue	Clock configuration for 16 kHz audio sampling rate needs verification in W2 to avoid WCKCFG errors.
Open-source tool(s) used	*STM32CubeIDE v1.16, STM32CubeU5 Firmware Package v1.3.0*.

SECTION 10 AI-ASSISTED ENGINEERING LOG

#	AI Tool Used	Task Given to AI (brief description of prompt)	AI Output Summary	Used In Deliverable?	Human Verification Performed
1	Gemini	Propose WBS for VUI Dispenser	Created 8-week breakdown for audio pipeline	Yes	Refined tasks to match STM32U5 I2S capability
2	Gemini	Generate technical phrasing for Section 1-2	Drafted academic objectives for Voice Activation on M33	Yes	Verified targets against current KWS Gantt chart
3	Claude	Debug	Main.c	No	Its still hasn’t worked

SECTION 11 LESSONS LEARNED

#	Lesson Learned — Write as a reusable engineering guideline with context, action, and reason
1	**[Baud Rate Stability]:** When initiating UART communication, always verify the core clock frequency in the .ioc file against the baud rate prescaler calculation. A mismatch between the system clock (HCLK) and the UART clock source is the most common cause of corrupted transmission; always verify using an oscilloscope if baud rate errors persist.
2	**[DSP Memory Management]:** In real-time Voice Activation, audio data processing must prioritize minimizing SRAM footprint. Always plan for "Ping-Pong" DMA buffering to allow the DSP pipeline to process one audio block while the I2S peripheral fills the next, preventing data drops without excessive CPU overhead.

SECTION 12 PLAN FOR NEXT WEEK

Define next week's tasks with enough specificity that a substitute engineer could continue the work without asking questions. Each task must reference the WBS category and have a clear, measurable expected output.

> 📝 **GUIDANCE:** *Carry-forward items (tasks not completed this week) must be listed first and marked 'CF' in the WBS Reference column. Explain why they were not completed and confirm they are a priority for next week.*

#	Next Task Description	WBS Reference	Priority	Expected Evidence / Deliverable
1	Configure I2S interface and GPDMA1 for audio streaming	WBS 3.1	High	Stable I2S clock and DMA transfer configuration
2	Develop Ping-Pong buffer logic in RAM for continuous audio	WBS 3.2	High	Buffer-swap proof-of-concept verified in debug log
3	Integrate CMSIS-DSP library to local environment and verify FFT	WBS 3.3	Medium	Successful function call of arm_rfft_fast_f32

SECTION 13 HISTORICAL NOTE FOR NEXT COHORT [MANDATORY]

Historical Note Category	Your Entry — Write for the next student who will do the same WBS task
If you were starting this WBS task today, what would you do first (before anything else)?	Immediately verify the hardware availability in the lab. If the requested ARM M7/FPGA hardware is unavailable, do not waste time waiting for procurement. Propose an architectural pivot to the STM32U5A9J-DK (Cortex-M33) immediately. This board is excellent for Voice Activation and has native DSP instructions (SIMD/MAC) via CMSIS-DSP which are sufficient for most undergraduate Edge AI/KWS research.
What was the most time-consuming problem you encountered?	The most time-consuming challenge was the initial phase of research and familiarization with the STM32 development environment (STM32CubeIDE). Navigating the complex clock tree configuration, dependency management for the STM32U5 series, and understanding the integration requirements for the ARM Cortex-M33 toolchain required a significant steep learning curve, which was compounded by the time spent reconciling the project mandate with actual laboratory hardware availability.
What mistake should the next student absolutely avoid?	Do not start your coding until you have successfully stabilized the UART and clock configuration on the board. Many students rush to AI modeling before validating the base firmware; if the serial communication is unstable or corrupted, you will never be able to debug the audio data flow later.
Which file, commit, code snippet, simulation script, or reference is the most useful starting point?	Refer to vui_baseline.ioc and the initial UART main.c implementation. This is the verified baseline; start from this configuration rather than generating a new project from scratch, as it already includes the correct pin mapping for the STM32U5A9J-DK.
What would you do differently if you had to repeat this week from scratch?	I would have presented the revised 8-week Gantt chart (focusing on STM32U5/Cortex-M33) to the industrial mentor in the first 2 hours of Week 1 to save the time I spent trying to align with the original, hardware-incompatible mandate.
What advice do you have that was not in the WBS or any reference document?	When doing Voice Activation (Keyword Spotting), do not just record clean studio-quality audio. Your model must be robust against the background noise of the physical dispenser (compressor/pump noise). Record the noise floor of your dispenser for 5 minutes and include it in your "background noise" class during Edge Impulse training, or your trigger word will fail in the real environment.

Note on Project Scope: The original project mandate was incompatible with the hardware inventory. After consultation with the mentor, we performed a proactive architectural pivot to the STM32U5 platform. This revised 8-week Gantt chart is the official baseline for this internship. Future cohorts should prioritize aligning the assigned project scope with available laboratory hardware during Week 0 to avoid schedule delays.

SECTION 14 SUPERVISOR REVIEW

Completed by the industrial mentor and/or academic supervisor after reviewing the WER. The student submits Rev 0.1; the supervisor annotates and returns it. The student incorporates feedback and resubmits as Rev 1.0.

Review Dimension	Supervisor Comment	Rating
Technical correctness	Are the measurements, calculations, and design decisions technically sound?	❑ Excellent ❑ Good ❑ Needs Work
Engineering reasoning	Is the student's explanation of cause-and-effect technically rigorous?	❑ Excellent ❑ Good ❑ Needs Work
Evidence quality	Are all results backed by traceable evidence files?	❑ Excellent ❑ Good ❑ Needs Work
Writing quality (for book)	Is the writing clear, precise, and at publishable quality?	❑ Excellent ❑ Good ❑ Needs Work
Literature references	Are the references relevant, correctly cited, and sufficient?	❑ Excellent ❑ Good ❑ Needs Work
Historical note quality	Is the historical note specific enough to be useful for the next cohort?	❑ Excellent ❑ Good ❑ Needs Work
Overall improvement from last week	Comment on growth trajectory	❑ Improved ❑ Same ❑ Declined

Field	Entry
Specific items to revise before Rev 1.0	…
Outstanding strength of this WER	…
Approval status	❑ Approved (Rev 1.0 accepted) ❑ Revise and resubmit ❑ Repeat test / simulation
Supervisor signature + date	…

SECTION 15 BOOK CHAPTER CONTRIBUTION STATEMENT [NEW — MANDATORY]

Book Chapter Contribution — Week 1 (Track C) Relevant book chapter title / section: Chapter 1 — DSP Architecture and Platform Adaptation for Smart Appliances

Contribution Statement:

This week established the foundational framework for deploying a Voice Activation system on the STM32U5 platform as an optimized DSP solution for automated dispensers. By successfully configuring the development environment and stabilizing the serial telemetry baseline.
This optimized Cortex-M33 approach serves as a critical case study in constraint-driven engineering and platform adaptation for smart appliance systems.

Furthermore, the initial literature review conducted this week bridges the gap between raw hardware constraints (such as clock tree stability for I2S audio) and the high-level Edge AI requirements necessary for reliable dispenser command detection. This phase establishes the reproducible baseline for future cohorts, specifically regarding the alignment of project scopes with actual laboratory resource limitations and the prioritization of robust communication telemetry before advancing to complex DSP pipeline integration.

Cross-track integration note (complete if your work this week connects to another track):

Integration Aspect	Description
How does this week's setup enable the DSP track?	The UART telemetry system provides the essential diagnostic window to monitor the performance of the MFCC feature extraction and keyword classification logic in real-time.
How does this week's hardware work interface with the Dispenser application?	The verified platform establishes the clock stability required for the I2S microphone interface, which is the primary sensor input for the voice-activated dispenser controls.
How does this week's work validate the overall system?	By validating the board functionality and clock tree early, we ensure that audio data acquisition in Week 2 will operate against a verified, noise-immune hardware baseline suitable for an appliance environment.